

LATTICES OF SUBVARIETIES AND OF LOGICS AS SUBLATTICES OF THE LATTICE OF TWO-OUT-OF-THREE CLASSES

RYOTA SHIBUSAWA

ABSTRACT. For a category C let $\text{Same}(C)$ be the complete lattice of classes of morphisms that contain the identities and are closed under composition and two-out-of-three. Every reflective subcategory $\mathcal{R}$ contributes the element $W_{\mathcal{R}}$ of maps inverted by the reflector. We study this assignment on the lattice $\text{Sub}(\mathcal{V})$ of subvarieties of a variety $\mathcal{V}$ of algebras. The meet of two subvarieties always goes to the join of the two classes, $W_{V \wedge V'} = W_V \vee W_{V'}$, because the reflector into $V \wedge V'$ is the composite of the two reflectors; for Grothendieck topologies the corresponding statement requires towers of plus constructions and can fail. The join of two subvarieties goes to the meet of the classes, $W_{V \vee V'} = W_V \cap W_{V'}$, whenever $\mathcal{V}$ is congruence distributive and congruence permutable, by Jónsson's lemma and a Chinese remainder argument; so for arithmetical varieties $V \mapsto W_V$ is a lattice embedding of $\text{Sub}(\mathcal{V})^{\text{op}}$ into $\text{Same}(\mathcal{V})$. Neither hypothesis can be dropped: the modular group of order 16 gives a counterexample in groups, and the three-element chain inside the four-element Boolean lattice gives one in bounded lattices with a constant. Through algebraizability the lattices of normal modal logics, of intermediate logics, of substructural logics over **FL** and of fuzzy logics are therefore sublattices of the sameness lattices of their algebras; the lattice of λ -theories embeds as a join-semilattice, and whether it embeds as a lattice is left open.

1. INTRODUCTION

A *sameness* on a category C is a class of morphisms containing the identities and closed under composition and two-out-of-three; the samenesses form a complete lattice $\text{Same}(C)$ under inclusion, with meets given by intersection and joins by closure [12]. Every functor F out of C gives a sameness $W_F = \{f : Ff \text{ invertible}\}$, and in earlier work the elements W_{a_J} contributed by the sheafification functors a_J of the Grothendieck topologies J on a small category were studied: $J \mapsto W_{a_J}$ preserves all meets, preserves binary joins for every finite category, and fails to preserve them on the poset of natural

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Corresponding author. E-mail: r-shibusawa@daiichi-koudai.ac.jp.

numbers [15, 16]. The positive results are proved by showing that alternating towers of plus constructions, or of separated quotients, connect a presheaf to its sheafification for the join topology inside the join of the two classes.

This paper carries out the same programme for the other classical lattice of reflective subcategories: the lattice $\text{Sub}(\mathcal{V})$ of subvarieties of a variety $\mathcal{V}$ of algebras. Each subvariety V is reflective, with reflector $a_V(A) = A/\theta_V(A)$ for the verbal congruence $\theta_V(A)$, and we write $W_V := W_{a_V} = \{f : a_V f \text{ is an isomorphism}\}$. The two lattice operations behave differently.

Theorem 1.1. *For every variety $\mathcal{V}$ and all subvarieties V, V' , $W_{V \wedge V'} = W_V \vee W_{V'}$ in $\text{Same}(\mathcal{V})$.*

Theorem 1.2. *For subvarieties V, V' of any variety, $W_{V \vee V'} = W_V \cap W_{V'}$ if and only if*

- (i) *every algebra of $V \vee V'$ is a subdirect product of its V -reflection and its V' -reflection, and*
- (ii) *no algebra of $V \vee V'$ has a proper subalgebra that is both V -dense and V' -dense, where a subalgebra is V -dense if its inclusion is inverted by the V -reflector.*

Condition (ii) holds whenever the verbal congruences $\theta_V(A)$ and $\theta_{V'}(A)$ permute for every $A \in V \vee V'$; together with (i) this says that A is the pullback of its V - and V' -reflections over its $V \wedge V'$ -reflection.

Corollary 1.3. *If $\mathcal{V}$ is congruence distributive and congruence permutable, then $W_{V \vee V'} = W_V \cap W_{V'}$ for all subvarieties V, V' , and $V \mapsto W_V$ is a lattice embedding $\text{Sub}(\mathcal{V})^{\text{op}} \hookrightarrow \text{Same}(\mathcal{V})$. Neither hypothesis can be omitted: the equality fails for the subvarieties $\text{var}(\mathbb{Z}_8)$ and $\text{var}(D_8)$ of groups (Example 4.3), and for the subvarieties $c = 0$ and $c = 1$ of bounded lattices with a constant c (Example 4.4); permutability, on the other hand, is not necessary (Example 4.5).*

Theorem 1.1 is elementary: because a subvariety is closed under quotients, the reflector into $V \wedge V'$ is the composite $a_{V'}a_V$, so the unit of $V \wedge V'$ is a composite of a V -unit and a V' -unit and the join theorem of [14] applies. The contrast with sheaf theory is the point: sheafification for J' does not preserve J -sheaves, the reflector into $J \vee J'$ is not a composite of the two, and the towers of [15, 16] are the substitute, which exists for finite categories and does not exist on $\mathbb{N}$. Theorem 1.2 is where the algebra enters. A map g of $V \vee V'$ lies in $W_{V \vee V'}$ only if it is an isomorphism, while $g \in W_V \cap W_{V'}$ only sees the two reflections of g ; so the question is whether the pair of reflectors $(a_V, a_{V'})$ reflects isomorphisms on $V \vee V'$, i.e. whether the “fracture” functor $V \vee V' \rightarrow V \times_{V \wedge V'} V'$ of Figure 2 is conservative. Condition (i) is injectivity, (ii) surjectivity. In a congruence distributive variety (i) holds for all pairs by Jónsson’s lemma [6], since the subdirectly irreducible algebras of $V \vee V'$ lie in $V \cup V'$; in a congruence permutable variety (ii) holds by a Chinese remainder argument. In an arithmetical variety every algebra of $V \vee V'$ is

therefore the pullback of its V - and V' -reflections over its $V \wedge V'$ -reflection, the algebraic analogue of a fracture square.

What is known. The reflector of a subvariety, verbal congruences, Jónsson's lemma and the Mal'cev condition for permutability are standard [3]. That a join of two subvarieties of a congruence permutable variety is the direct product of the two when they are disjoint is due to Jónsson and Tsinakis; joins that are subdirect products are studied by Kowalski and Paoli [8]. The lattice of samenesses, the join theorem and the results for Grothendieck topologies are from [12, 14, 15, 16].

What is new. Theorems 1.1 and 1.2, the three examples, and the consequences for logic (Section 5): since the lattice of axiomatic extensions of an algebraizable logic is dually isomorphic to the subvariety lattice of its equivalent algebraic semantics [1], the lattices of normal modal logics, of intermediate logics, of substructural logics over **FL** and of the usual fuzzy logics are sublattices of the sameness lattices of modal algebras, Heyting algebras, **FL**-algebras and their subvarieties (Corollary 5.1); the lattice of λ -theories, which is isomorphic to the lattice of equational theories of lambda abstraction algebras [10], embeds as a join-semilattice (Corollary 5.3), and the question whether it embeds as a lattice is left open, as lambda abstraction algebras satisfy no non-trivial congruence lattice identity [9].

2. PRELIMINARIES

2.1. Samenesses. For a category C , $\text{Same}(C)$ is the set of samenesses ordered by inclusion; it is a complete lattice, $U \wedge U' = U \cap U'$, and $U \vee U'$ is the smallest sameness containing $U \cup U'$ [12]. For a functor $P: C \rightarrow C$ put $W_P = \{f : Pf \text{ invertible}\}$; it is a sameness containing all isomorphisms.

Theorem 2.1 (join theorem, [14, Thm. 4.1]). *Let U, V be samenesses on C whose join contains the isomorphisms, and let $P: C \rightarrow C$ be a functor with a natural transformation $\pi: \text{id} \Rightarrow P$ such that $\pi_X \in U \vee V$ for all X and P inverts U and V . Then $U \vee V = W_P$.*

Proof. $W_P \supseteq U \vee V$ since W_P is a sameness containing U and V . If Pf is invertible then $Pf \in U \vee V$, so the naturality square $\pi_Y f = Pf \pi_X$ has three sides in $U \vee V$, hence the fourth. $\square$

2.2. Subvarieties and verbal congruences. Let $\mathcal{V}$ be a variety of algebras of a fixed signature and $\text{Sub}(\mathcal{V})$ its lattice of subvarieties; $V \wedge V' = V \cap V'$ and $V \vee V'$ is the variety generated by $V \cup V'$. For $A \in \mathcal{V}$ and $V \in \text{Sub}(\mathcal{V})$ the *verbal congruence* $\theta_V(A)$ is the smallest congruence of A whose quotient lies in V ; it is the congruence generated by the pairs $(s(\bar{a}), t(\bar{a}))$ for the identities $s = t$ of V and tuples $\bar{a}$ of A , and a homomorphism $f: A \rightarrow B$ satisfies $f(\theta_V(A)) \subseteq \theta_V(B)$. Thus $a_V(A) = A/\theta_V(A)$ is a reflector of $\mathcal{V}$ onto V with unit the quotient map $u_A: A \rightarrow a_V A$, and we put

$$W_V := \{f \in \text{Mor}(\mathcal{V}) : a_V f \text{ is an isomorphism}\} \in \text{Same}(\mathcal{V}).$$

If $V \subseteq V'$ then $a_V = a_V a_{V'}$ and $W_{V'} \subseteq W_V$. The map $V \mapsto W_V$ is an order-embedding $\text{Sub}(\mathcal{V})^{\text{op}} \hookrightarrow \text{Same}(\mathcal{V})$: if $V \not\subseteq V'$, pick $A \in V \setminus V'$; its V' -unit is in $W_{V'}$, not an isomorphism, and a_V sends it to itself (as $a_V A = A$ and $a_V(A/\theta_{V'}(A)) = A/\theta_{V'}(A)$ because both lie in V , which is closed under quotients), so it is not in W_V . We write $\text{Mor}(\mathcal{V})$ for the class of homomorphisms between algebras of $\mathcal{V}$.

Lemma 2.2. *For $V, V' \in \text{Sub}(\mathcal{V})$ and $A \in \mathcal{V}$ the composite $A \rightarrow A/\theta_V(A) \rightarrow (A/\theta_V(A))/\theta_{V'}(A/\theta_V(A))$ is the $V \wedge V'$ -unit of A ; that is, $a_{V \wedge V'} = a_{V'} a_V$ and $\theta_{V \wedge V'}(A) = \theta_V(A) \vee \theta_{V'}(A)$.*

Proof. The target lies in V' by construction and in V because V is closed under quotients. Any homomorphism $A \rightarrow C$ with $C \in V \cap V'$ factors through $A/\theta_V(A)$ since $C \in V$, and the factor through $(A/\theta_V(A))/\theta_{V'}$ since $C \in V'$. By the correspondence theorem the kernel of the composite is $\theta_V(A) \vee \theta_{V'}(A)$. $\square$

3. MEETS OF SUBVARIETIES

Proof of Theorem 1.1. Apply Theorem 2.1 with $U = W_V$, $V = W_{V'}$ (the join contains the isomorphisms because each does), $P = a_{V \wedge V'}$ and π the unit. By Lemma 2.2, π_A is the composite of the V -unit of A , which lies in W_V because a_V of a unit is an isomorphism, and the V' -unit of $A/\theta_V(A)$, which lies in $W_{V'}$; so $\pi_A \in W_V \vee W_{V'}$. Since $a_{V \wedge V'} = a_{V \wedge V'} a_V = a_{V \wedge V'} a_{V'}$, the functor $a_{V \wedge V'}$ inverts W_V and $W_{V'}$. Hence $W_V \vee W_{V'} = W_{a_{V \wedge V'}} = W_{V \wedge V'}$. $\square$

$$\begin{array}{c}
 A \xrightarrow{u_A \in W_V} A/\theta_V(A) \xrightarrow{u' \in W_{V'}} A/(\theta_V(A) \vee \theta_{V'}(A)) = a_{V \wedge V'} A \\
 \searrow \hspace{10em} \nearrow \\
 \hspace{10em} \pi_A
 \end{array}$$

FIGURE 1. The unit of $V \wedge V'$ is a composite of two units (Lemma 2.2); this single step replaces the towers needed for Grothendieck topologies.

The argument uses only that each subvariety is closed under the reflections into the other; it holds verbatim for reflective subcategories $\mathcal{R}, \mathcal{R}'$ of any category with $a_{\mathcal{R}'}(\mathcal{R}) \subseteq \mathcal{R}$, where then $a_{\mathcal{R} \cap \mathcal{R}'} = a_{\mathcal{R}'} a_{\mathcal{R}}$ and $W_{\mathcal{R}} \vee W_{\mathcal{R}'} = W_{\mathcal{R} \cap \mathcal{R}'}$. Full subcategories of sheaves do not have this property: for two Grothendieck topologies J, J' the J' -sheafification of a J -sheaf need not be a J -sheaf, $a_{J \vee J'}$ is not the composite $a_{J'} a_J$, and the join $W_{a_J} \vee W_{a_{J'}}$ is the class $W_{a_{J \vee J'}}$ only when a tower of alternating constructions reaches the join sheafification, which it does for finite categories [16] and does not on the poset $\mathbb{N}$ [15, Thm. 7.1].

4. JOINS OF SUBVARIETIES

The inclusion $W_{V \vee V'} \subseteq W_V \cap W_{V'}$ always holds, since $a_V = a_V a_{V \vee V'}$ and $a_{V'} = a_{V'} a_{V \vee V'}$. For the converse we reduce to maps inside $V \vee V'$.

Lemma 4.1. *$W_{V \vee V'} = W_V \cap W_{V'}$ holds if and only if every homomorphism $g: A \rightarrow B$ with $A, B \in V \vee V'$ for which $a_V g$ and $a_{V'} g$ are isomorphisms is itself an isomorphism.*

Proof. If $f \in W_V \cap W_{V'}$, then $g := a_{V \vee V'} f$ is a map in $V \vee V'$ with $a_V g = a_V f$ and $a_{V'} g = a_{V'} f$ isomorphisms; if the condition holds, g is an isomorphism and $f \in W_{V \vee V'}$. Conversely, a map g in $V \vee V'$ has $a_{V \vee V'} g = g$, so $g \in W_{V \vee V'}$ iff g is an isomorphism. $\square$

Definition 4.2. A subalgebra $C \subseteq B$ is *V-dense* if the inclusion lies in W_V , i.e. if C meets every class of $\theta_V(B)$ and $\theta_V(B) \cap C^2 = \theta_V(C)$.

Proof of Theorem 1.2. Suppose (i) and (ii) and let $g: A \rightarrow B$ be a map in $V \vee V'$ with $a_V g, a_{V'} g$ isomorphisms; by Lemma 4.1 we must show that g is bijective. If $g(x) = g(y)$ then $x \theta_V(A) y$ and $x \theta_{V'}(A) y$ because $a_V g$ and $a_{V'} g$ are injective, so $x = y$ by (i): g is injective. Its image $C := g(A)$ is a subalgebra of B with $C \cong A$, and the inclusion $C \hookrightarrow B$ is inverted by a_V and $a_{V'}$ (it is g composed with an isomorphism); so C is V -dense and V' -dense and $C = B$ by (ii).

Conversely suppose $W_{V \vee V'} = W_V \cap W_{V'}$. If (i) fails for some $A \in V \vee V'$, let $\theta := \theta_V(A) \cap \theta_{V'}(A) \neq \Delta$ and $g: A \rightarrow B := A/\theta$ the quotient map; $B \in V \vee V'$ as a subdirect product of $A/\theta_V(A) \in V$ and $A/\theta_{V'}(A) \in V'$. By the correspondence theorem $\theta_V(B) = (\theta_V(A) \vee \theta)/\theta = \theta_V(A)/\theta$, so $a_V g$ is the identity of $A/\theta_V(A)$, and likewise for V' ; thus $g \in W_V \cap W_{V'}$, but g is not injective, contradicting Lemma 4.1. If (ii) fails, a V -dense and V' -dense proper subalgebra $C \subsetneq B$ with $B \in V \vee V'$ gives an inclusion in $W_V \cap W_{V'}$ which is not an isomorphism, again contradicting Lemma 4.1.

Finally let the verbal congruences permute on every algebra of $V \vee V'$ and let $C \subseteq B$ be V -dense and V' -dense, $B \in V \vee V'$; we show $C = B$. Write $\theta_V, \theta_{V'}$ for the verbal congruences of B . Given $b \in B$, density gives $c_1, c_2 \in C$ with $c_1 \theta_V b$ and $c_2 \theta_{V'} b$. Then c_1 and c_2 are related by $\theta_V \vee \theta_{V'} = \theta_{V \wedge V'}$ (Lemma 2.2), hence, as $a_{V \wedge V'} = a_{V'} a_V$ inverts the inclusion, by $\theta_{V \wedge V'}(C) = \theta_V(C) \vee \theta_{V'}(C)$ inside C ; by permutability in C (also an algebra of $V \vee V'$) there is $c \in C$ with $c_1 \theta_V(C) c \theta_{V'}(C) c_2$. Then $c \theta_V c_1 \theta_V b$ and $c \theta_{V'} c_2 \theta_{V'} b$, so $c(\theta_V \cap \theta_{V'}) b$ and $c = b$ by (i) for B . $\square$

Proof of Corollary 1.3. Let $\mathcal{V}$ be congruence distributive and congruence permutable. The verbal congruences permute, so (ii) holds by Theorem 1.2. For (i), let $A \in V \vee V'$. By Jónsson's lemma [6] every subdirectly irreducible algebra of $V \vee V'$ lies in $HSP_u(V \cup V')$; an ultraproduct of algebras each in V or in V' lies in V or in V' (one of the two index sets belongs to the ultrafilter), and V, V' are closed under H and S , so the subdirectly irreducibles of $V \vee V'$ lie in $V \cup V'$. Writing A as a subdirect product of subdirectly

$$\begin{array}{ccc}
A & \xrightarrow{\quad} & A/\theta_V(A) \\
\downarrow & \lrcorner & \downarrow \\
A/\theta_{V'}(A) & \longrightarrow & A/(\theta_V(A) \vee \theta_{V'}(A))
\end{array}$$

FIGURE 2. The fracture square of $A \in V \vee V'$. Condition (i) of Theorem 1.2 is injectivity of the comparison map $A \rightarrow A/\theta_V \times_{A/(\theta_V \vee \theta_{V'})} A/\theta_{V'}$, and permutability of $\theta_V(A)$, $\theta_{V'}(A)$ is its surjectivity; the square is a pullback for every A in an arithmetical variety.

irreducible quotients A/κ_i , each κ_i contains $\theta_V(A)$ or $\theta_{V'}(A)$, hence contains $\theta_V(A) \cap \theta_{V'}(A)$; as $\bigcap_i \kappa_i$ is the identity, so is $\theta_V(A) \cap \theta_{V'}(A)$, which is (i). Theorem 1.2 gives $W_{V \vee V'} = W_V \cap W_{V'}$, and with Theorem 1.1 and the order-embedding of Section 2 the map $V \mapsto W_V$ is a lattice embedding. The examples below show that the hypotheses cannot be dropped and that permutability is not necessary. $\square$

Example 4.3 (permutable, not distributive). In the variety of groups let $V = \text{var}(\mathbb{Z}_8)$, the abelian groups of exponent dividing 8, and $V' = \text{var}(D_8)$, the variety generated by the dihedral group of order 8. Let $M_{16} = \langle a, b \mid a^8 = b^2 = 1, bab = a^5 \rangle$ be the modular group of order 16. It lies in $V \vee V'$: in $\mathbb{Z}_8 \times D_8$, with r a rotation of order 4 and s a reflection, the subgroup K generated by $x = (1, r)$ and $y = (0, s)$ contains $x \cdot yxy = (1, r)(1, r^{-1}) = (2, 1)$, hence $x^2(2, 1)^{-1} = (0, r^2)$ and $n := x^4(0, r^2) = (4, r^2)$, a central element of order 2. In $K/\langle n \rangle$ the image of x has order 8 (as $x^4 = (4, 1) \notin \langle n \rangle$), the image of y has order 2, and $yxy = (1, r^{-1}) = (5, r)(4, r^2) = x^5 n \equiv x^5$. So $K/\langle n \rangle$, which has order 16 because $K = \{(i, r^j s^k) : i \equiv j \pmod{2}\}$ has order 32, is generated by two elements satisfying the defining relations of M_{16} , whence $K/\langle n \rangle \cong M_{16}$ and $M_{16} \in HS(\mathbb{Z}_8 \times D_8) \subseteq V \vee V'$. The group M_{16} is subdirectly irreducible, its unique minimal normal subgroup being $\langle a^4 \rangle$, and it is in neither V (non-abelian) nor V' (exponent 8, while V' has exponent 4). The verbal subgroups are $\theta_V(M_{16}) = M'_{16} M_{16}^8 = \langle a^4 \rangle$ and $\theta_{V'}(M_{16}) \supseteq M_{16}^4 = \langle a^4 \rangle$, with equality because $M_{16}/\langle a^4 \rangle \cong \mathbb{Z}_4 \times \mathbb{Z}_2 \in V'$. Thus condition (i) fails for $A = M_{16}$, and by Theorem 1.2 the quotient map $M_{16} \rightarrow \mathbb{Z}_4 \times \mathbb{Z}_2$ lies in $W_V \cap W_{V'}$ but not in $W_{V \vee V'}$. Groups are congruence permutable, so it is distributivity that is missing.

Example 4.4 (distributive, not permutable). Let $\mathcal{V}$ be the variety of bounded lattices $(L, \wedge, \vee, 0, 1)$ with an additional constant c , which is congruence distributive, and let V and V' be the subvarieties defined by $c = 0$ and by $c = 1$. Let $B = \mathbf{2} \times \mathbf{2}$ with $c = (0, 1)$; it is the product of a V -algebra and a V' -algebra, so $B \in V \vee V'$, and $\theta_V(B) = \text{Cg}((0, 1), (0, 0))$ is the kernel of the first projection while $\theta_{V'}(B) = \text{Cg}((0, 1), (1, 1))$ is the kernel of the second. Let $C = \{(0, 0), (0, 1), (1, 1)\}$ be the three-element

chain, a subalgebra of B containing c . Then $\theta_V(C)$ identifies $(0,0)$ with $(0,1)$ and $\theta_{V'}(C)$ identifies $(0,1)$ with $(1,1)$, so the inclusion $g: C \hookrightarrow B$ induces bijections $C/\theta_V(C) \rightarrow B/\theta_V(B)$ and $C/\theta_{V'}(C) \rightarrow B/\theta_{V'}(B)$, both two-element lattices, while g is not surjective: C is a V -dense and V' -dense proper subalgebra, (ii) fails, and $g \in W_V \cap W_{V'} \setminus W_{V \vee V'}$. Condition (i) holds for all algebras of $V \vee V'$ by distributivity. The verbal congruences of C do not permute: $(1,1)$ and $(0,0)$ are related by $\theta_V(C) \vee \theta_{V'}(C)$ but not by $\theta_V(C) \circ \theta_{V'}(C)$, and C is a proper subalgebra of its pullback, which is B .

Example 4.5 (non-permuting verbal congruences, equality holds). Let $\mathcal{V}$ be the variety of mono-unary algebras (A, f) with $f^4 = f^2$, and $V: f^2x = x$, $V': f^2x = fx$. The identities $f^p x = f^q x$ true in both are those with $p \equiv q \pmod{2}$ and $p, q \geq 1$ (or $p = q$), so $V \vee V'$ is defined by $f^3x = fx$. An algebra B of $V \vee V'$ consists of its *core* $f(B)$, on which f is an involution, and *hairs* $x \notin f(B)$ with $f(x) \in f(B)$. The congruence $\theta_V(B)$, generated by the pairs (f^2x, x) , identifies each hair x with the core element f^2x and nothing else; $\theta_{V'}(B)$, generated by the pairs (f^2x, fx) , collapses each 2-cycle of the core to a point and nothing else. Hence $\theta_V(B) \cap \theta_{V'}(B) = \Delta$, which is (i). For (ii), let $C \subseteq B$ be V -dense and V' -dense. Density for V means that $C/\theta_V(C)$, which is the core of C , maps onto the core of B , so $C \supseteq f(B)$; density for V' means that the hairs of C are all the hairs of B , since hairs are not identified by $\theta_{V'}$. Thus $C = B$, and $W_{V \vee V'} = W_V \cap W_{V'}$ by Theorem 1.2. Yet the verbal congruences do not permute on the three-element algebra $A = \{0, 1, 2\}$ with $f(0) = 1$, $f(1) = 2$, $f(2) = 1$: $\theta_V(A)$ identifies 0 with 2, $\theta_{V'}(A)$ identifies 1 with 2, and $(1, 0) \in \theta_{V'} \circ \theta_V$ but $(1, 0) \notin \theta_V \circ \theta_{V'}$; A is a proper subalgebra of its pullback $A/\theta_V \times A/\theta_{V'}$, which has four elements and in which A is not V' -dense.

Remark 4.6. Condition (i) for all $A \in V \vee V'$ says that $V \vee V'$ consists of subdirect products of algebras of V and of V' , which Kowalski and Paoli [8] study for disjoint subvarieties; in the congruence distributive case it holds for all pairs by Jónsson's lemma. Condition (ii) is a statement about dense subalgebras which permutability implies but which, by Example 4.5, is weaker. We do not know a characterization of the pairs (V, V') with $W_{V \vee V'} = W_V \cap W_{V'}$ in terms of the lattice $\text{Sub}(\mathcal{V})$ alone.

5. LATTICES OF LOGICS AND OF λ -THEORIES

For an algebraizable logic $\mathcal{L}$ with equivalent algebraic semantics the variety $\mathcal{V}_{\mathcal{L}}$, the lattice of axiomatic extensions of $\mathcal{L}$ is dually isomorphic to $\text{Sub}(\mathcal{V}_{\mathcal{L}})$, an extension L corresponding to the subvariety $\mathcal{V}_L$ of algebras validating its theorems [1]. Write $W_L := W_{\mathcal{V}_L}$, the class of homomorphisms of $\mathcal{V}_{\mathcal{L}}$ -algebras inverted by the reflection into $\mathcal{V}_L$; $L \leq L'$ gives $W_L \subseteq W_{L'}$.

Corollary 5.1. *Let $\mathcal{L}$ be an algebraizable logic whose equivalent algebraic semantics $\mathcal{V}_{\mathcal{L}}$ is congruence distributive and congruence permutable. Then $L \mapsto W_L$ is a lattice embedding of the lattice of axiomatic extensions of $\mathcal{L}$*

into $\text{Same}(\mathcal{V}_{\mathcal{L}})$: $W_{L \wedge L'} = W_L \cap W_{L'}$ and $W_{L \vee L'} = W_L \vee W_{L'}$. This applies to

- (1) the lattice $\text{NExt}(\mathbf{K})$ of normal modal logics, with $\mathcal{V}_{\mathcal{L}}$ the modal algebras [4];
- (2) the lattice of intermediate logics, with Heyting algebras;
- (3) the lattice of substructural logics over the full Lambek calculus **FL**, with **FL**-algebras, and its intervals of fuzzy logics over **MTL**, **BL**, Łukasiewicz and Gödel logic, with **MTL**-, **BL**-, **MV**- and Gödel algebras.

Proof. Theorem 1.1 and Corollary 1.3 with $\mathcal{V} = \mathcal{V}_{\mathcal{L}}$, using the dual isomorphism $L \mapsto \mathcal{V}_L$; the meet of logics is the join of varieties and conversely. Each of the varieties listed has a lattice reduct, hence is congruence distributive, and a Mal'cev term: for modal algebras, which have a Boolean reduct, $x \oplus y \oplus z$ with $\oplus$ symmetric difference; for Heyting algebras and their expansions the standard term; for **FL**-algebras $p(x, y, z) = x/(z \setminus y \wedge 1) \wedge z/(x \setminus y \wedge 1)$ [7, 2, 5], which also serves all their subvarieties. $\square$

Remark 5.2. In the framework of [13], where $(\text{Same}(C), \subseteq)$ is read as a Kripke frame whose worlds are the levels of sameness, Corollary 5.1 says that the lattice of normal modal logics sits inside the frame of modal algebras as a sublattice: the logics themselves are levels of sameness for the algebras of the base logic.

Corollary 5.3. Let $\lambda\mathcal{T}$ be the lattice of λ -theories and **LAA** the variety of lambda abstraction algebras. The assignment $T \mapsto W_T := W_{\mathcal{V}_T}$, where $\mathcal{V}_T$ is the subvariety generated by the term algebra Λ/T , is an injective map $\lambda\mathcal{T} \rightarrow \text{Same}(\mathbf{LAA})$ preserving order and binary joins: $W_{T \vee T'} = W_T \vee W_{T'}$; and always $W_{T \wedge T'} \subseteq W_T \cap W_{T'}$.

Proof. By [10] the lattice $\lambda\mathcal{T}$ is isomorphic to the lattice of equational theories of **LAA**, i.e. dually isomorphic to $\text{Sub}(\mathbf{LAA})$, by $T \mapsto \mathcal{V}_T$; joins of theories correspond to meets of varieties, and Theorem 1.1 applies. $\square$

Question 5.4. Is $T \mapsto W_T$ a lattice embedding $\lambda\mathcal{T} \hookrightarrow \text{Same}(\mathbf{LAA})$, i.e. does $W_{T \wedge T'} = W_T \cap W_{T'}$ hold for all λ -theories? By Theorem 1.2 this requires every subdirectly irreducible algebra of $\mathcal{V}_T \vee \mathcal{V}_{T'}$ to satisfy T or T' . Corollary 1.3 does not apply: lambda abstraction algebras satisfy no non-trivial lattice identity in their congruence lattices [9], and $\lambda\mathcal{T}$ itself is not modular [11].

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$$\begin{array}{ccc}
\text{NExt}(\mathbf{K}) & \xrightarrow{\text{lattice}} & \text{Same}(\mathbf{MA}) \\
\lambda\mathcal{T} & \xrightarrow[\vee\text{-semilattice}]{} & \text{Same}(\mathbf{LAA}) \\
\text{Ext}(\mathbf{FL}) & \xrightarrow{\text{lattice}} & \text{Same}(\mathbf{FL}) \\
\text{Top}(S) & \xrightarrow[\text{lattice}, S \text{ finite}]{} & \text{Same}(\text{Presh}(S))
\end{array}$$

FIGURE 3. Lattices of theories inside sameness lattices: logics (Corollary 5.1), λ -theories (Corollary 5.3), Grothendieck topologies [16].

DECLARATIONS

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DAIICHI INSTITUTE OF TECHNOLOGY

Email address: `r-shibusawa@daiichi-koudai.ac.jp`